

Multiple Choice (10 marks)

1. Which of the following sets has the greatest cardinality?
 - (a) The set of all finite subsets of $\mathbb{R}$
 - (b) $\mathbb{R}$
 - (c) The set of all infinite subsets of $\mathbb{R}$
 - (d) The set of all functions from $\mathbb{Z}$ to $\mathbb{Z}$
 - (e) The set of all functions from $\{0, 1\}$ to $\mathbb{R}$
2. There are 25 suitcases, 5 of which are damaged. Three suitcases are selected at random. What is the probability that exactly 2 are damaged?
 - (a) $2/50$
 - (b) $1/25$
 - (c) $1/30$
 - (d) $3/75$
 - (e) $4/100$
3. A grocery store sells a bag of 5 lemons for \$2.00. What is the unit cost of each lemon in the bag?
 - (a) \$1.50
 - (b) \$0.50
 - (c) \$0.20
 - (d) \$0.60
 - (e) \$0.10
4. What number makes the equation $35 / ? = 7$ true?
 - (a) 5
 - (b) 25
 - (c) 7
 - (d) 6
 - (e) 10
5. Compute the product in the given ring. $(12)(16)$ in $\mathbb{Z}_{24}$
 - (a) 24
 - (b) 28

- (c) 20
- (d) 16
- (e) 8

True / False (10 marks)

Answer True or False

6. True or False: The cost of one pencil is 66 cents when eleven pencils cost the same as three pens, which total \$9.24 for seven pens.
7. True or False: The solution to the system of equations is given by the values $x = 1$, $y = 3$, and $z = -1$.
8. True or False: There are 240 ways to divide a set of 5 elements into 2 non-empty ordered subsets.
9. True or False: The expression $1990 \times 1991 - 1989 \times 1990$ simplifies to 3970.
10. True or False: The value of $y(4)$ for the differential equation $dy/dt =$ with the initial condition $y(1) = 1$ is approximately 7.789.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze the division of 2,314 by 4, providing the quotient and remainder, and discuss the implications of these results in a broader mathematical context.
12. Determine the smallest positive integer that satisfies the following modular conditions: it leaves a remainder of 5 when divided by 8, a remainder of 1 when divided by 3, and a remainder of 7 when divided by 11. Explain your reasoning and the methods used to arrive at your solution.

End of Paper